

Exact Cosparse Projection for a Complete-Graph Tight Frame

A partial result for arXiv:1207.2456

FA/Banach run `fa_banach_001`, lane 7

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Status

candidate partial result -- likely valid pending human review. This packet gives a new exact polynomial-time projection algorithm for a structured incidence operator and for a full-rank redundant tight-frame completion. It does not settle projection or approximation for arbitrary analysis operators.

1 Source question

Giryes, Nam, Elad, Gribonval, and Davies study greedy-like algorithms for the cosparse analysis model [1]. Their convergence theorems assume an optimal or near-optimal cosupport-selection oracle. In the conclusion, PDF page 26, they ask for which structured operators such an efficient oracle exists and single out tight-frame operators for special attention.

performance.

We are aware that there are still some open questions in this work and we leave them for future research. This should deal with following:

- Our work assumed the existence of a procedure that finds a cosupport that implies a near optimal projection with a constant C_ℓ . Two examples for optimal cosupport selection schemes were given. However, the existence of an optimal or a near optimal scheme for a general operator is still an open question. The question is: for which types of Ω and values of C_ℓ we can find an efficient procedure that implies a near optimal projection.
- As we have seen in the simulations, the thresholding procedure, though not near optimal with the theorems required constants, provides good reconstruction results. A theoretical study of the analysis greedy-like techniques with this cosupport selection scheme is required.
- A family of analysis dictionaries that deserves a special attention is the family of tight frame operators. In synthesis, there is a parallel between the guarantees of ℓ_1 -synthesis and the greedy like algorithms. The fact that a guarantee with a tight frame Ω exists for ℓ_1 -analysis encourage us to believe that similar guarantees exist also for the analysis greedy-like techniques.

Figure 1: The source paper's open projection question and tight-frame direction, arXiv:1207.2456, PDF page 26.

The source defines an ℓ -cosparse vector by requiring at least ℓ zero coefficients in Ωx . A cosupport selector has constant C_ℓ if its squared projection error is at most C_ℓ times the optimal squared error. Exact projection means $C_\ell = 1$.

2 Candidate theorem

Let $B \in \mathbb{R}^{\binom{n}{2} \times n}$ be an oriented edge–vertex incidence matrix of the complete graph K_n : the row indexed by $i < j$ is $e_i^T - e_j^T$. Set

$$Z_B(x) := \#\{(i, j) : 1 \leq i < j \leq n, x_i = x_j\}$$

and, for $0 \leq r \leq \binom{n}{2}$,

$$F_r(z) := \min\{\|z - x\|_2^2 : Z_B(x) \geq r\}.$$

Theorem 1 (Complete-graph incidence projection). *For every $z \in \mathbb{R}^n$ and every r , the value $F_r(z)$ and a minimizing vector can be computed exactly using $O(n^2 \binom{n}{2}) = O(n^4)$ arithmetic operations and $O(n \binom{n}{2}) = O(n^3)$ storage. The algorithm also returns a cosupport Λ such that the minimizer is $Q_\Lambda z$. Hence its near-optimality constant is $C_r = 1$.*

Now append one row:

$$\Omega = \begin{bmatrix} B \\ \mathbf{1}^T \end{bmatrix}.$$

Since $B^T B = nI - \mathbf{1}\mathbf{1}^T$, this is a full-rank redundant tight-frame analysis operator:

$$\Omega^T \Omega = nI.$$

Equivalently, $n^{-1/2}\Omega$ is Parseval; this common nonzero row scaling does not change any zero coefficient or cosupport subspace. Let

$$G_r(z) := \min\{\|z - x\|_2^2 : \#\{k : (\Omega x)_k = 0\} \geq r\}.$$

Theorem 2 (Tight completion). *Write $\bar{z} = n^{-1}\mathbf{1}^T z$, put $F_{-1}(z) = 0$, and regard $F_r(z) = +\infty$ when $r > \binom{n}{2}$. Then $G_0(z) = 0$ and, for $1 \leq r \leq \binom{n}{2} + 1$,*

$$G_r(z) = \min \{F_r(z), F_{r-1}(z) + n\bar{z}^2\}.$$

Both branches, a minimizing vector, and an exact cosupport are computed by the dynamic program of Theorem 1. Thus the tight-frame operator has an efficient exact projection oracle with $C_r = 1$ for every target cosparsity.

3 Proof intuition

For K_n , every zero analysis coefficient says that two coordinates agree. Thus a candidate vector is described by a partition of the coordinates into equality blocks, and a block of size s contributes $\binom{s}{2}$ zeros. For a fixed partition, the closest vector replaces the data on each block by their mean. The key one-dimensional observation is that after sorting the data, an optimal partition with any prescribed multiset of block sizes can be taken to consist of consecutive intervals. The projection problem therefore becomes a finite segmentation dynamic program with an extra state recording the accumulated number of equal pairs.

The appended all-ones row creates only two cases. If it is nonzero, at least r pair-difference rows must vanish. If it is zero, only $r - 1$ pair rows are needed, and the projected vector must have mean zero. Enforcing mean zero shifts every block mean by the same global mean and adds exactly $n\bar{z}^2$ to the squared error.

4 Proofs

Lemma 1 (Contiguous blocks). *Let $y_1 \leq \dots \leq y_n$ and fix positive block sizes $s_1, \dots, s_m$ summing to n . Among all partitions of the data into blocks with this multiset of sizes, one minimizing the within-block sum of squares can be chosen so that every block is a consecutive interval in the sorted order.*

Proof. Take an optimal partition, let its block means be $\mu_1 \leq \dots \leq \mu_m$, and relabel the corresponding sizes in that order. Temporarily keep these means fixed. Assigning the sorted data to means with capacities s_j has cost matrix $c_{ij} = (y_i - \mu_j)^2$. For $i < i'$ and $j < j'$,

$$c_{ij} + c_{i'j'} \leq c_{ij'} + c_{i'j},$$

because the right side minus the left side is $2(y_{i'} - y_i)(\mu_{j'} - \mu_j) \geq 0$. Hence crossing assignments may be uncrossed without increasing cost. Repeated uncrossing gives the consecutive capacity-respecting assignment. Replacing each fixed mean by the mean of its new block can only decrease the cost. The original partition was optimal, so the resulting consecutive partition is also optimal and has the same block sizes. $\square$

Proof of Theorem 1. Any x determines equality blocks. Replacing its value on each block by the mean of the corresponding entries of z cannot increase the error and cannot decrease the number of zero pair differences; coincident new means merely merge blocks. By Lemma 1, an optimum can therefore be chosen as consecutive blocks after sorting z .

Let $y_1 \leq \dots \leq y_n$ be the sorted data and define the interval cost

$$E(a, b) = \sum_{i=a}^b (y_i - \bar{y}_{a:b})^2.$$

Prefix sums of y_i and y_i^2 evaluate every $E(a, b)$ in constant time. Let $D(i, q)$ be the minimum cost of partitioning $y_1, \dots, y_i$ into consecutive blocks contributing exactly q equal pairs. Initialize $D(0, 0) = 0$ and all other states to $+\infty$. The recurrence is

$$D(i, q) = \min_{0 \leq j < i} \left\{ D\left(j, q - \binom{i-j}{2}\right) + E(j+1, i) \right\},$$

where states with an invalid second index are omitted. Therefore

$$F_r(z) = \min_{q \geq r} D(n, q).$$

There are $O(n \binom{n}{2})$ states and at most n transitions per state. Backpointers recover the intervals and their means. Taking every within-block edge as Λ gives precisely the subspace of block-constant vectors, so the recovered minimizer equals $Q_\Lambda z$. $\square$

Proof of Theorem 2. The Gram identity follows from the complete-graph Laplacian: $B^T B = nI - \mathbf{1}\mathbf{1}^T$. Moreover,

$$\#\{k : (\Omega x)_k = 0\} = Z_B(x) + \mathbf{1}_{\{\mathbf{1}^T x = 0\}}.$$

Thus the feasible set at level $r \geq 1$ is the union of the level- r incidence set and the intersection of the level- $(r-1)$ incidence set with $\mathbf{1}^\perp$.

For a fixed equality partition, its block-constant subspace contains $\mathbf{1}$. The unconstrained projection uses block means and preserves the global sum of z . Its projection onto the additional

hyperplane $\mathbf{1}^\perp$ subtracts $\bar{z}\mathbf{1}$. Orthogonality gives the extra squared cost $\|\bar{z}\mathbf{1}\|_2^2 = n\bar{z}^2$, independent of the partition. Minimizing the two branches gives the displayed formula. The second branch is recovered from an optimal level- $(r - 1)$ incidence partition followed by subtraction of $\bar{z}\mathbf{1}$; the equality pattern is unchanged. $\square$

Corollary 1 (Clique unions and the source algorithms). *For an incidence operator of a disjoint union of cliques, compute a local table on each clique and convolve the tables over the total cosparsity budget. Appending the component-sum row on each clique gives the analogous tight completion locally; nonzero componentwise row scaling normalizes the block frame bounds without changing any cosupport. These projections remain polynomial-time and exact.*

In particular, for the single-clique tight operator in Theorem 2, every projection constant assumed in the source paper’s AIHT, AHTP, ACoSaMP, and ASP theorems is algorithmically available with value one. Their recovery conclusions therefore apply under their original Ω -RIP and noise hypotheses without an unavailable projection oracle.

5 Exact verification

The included script `code/verify_complete_graph_projection.py` implements both dynamic programs using rational arithmetic and compares them with exhaustive enumeration of all set partitions. The default run checks every vector in $\{-1, 0, 1\}^n$ for $1 \leq n \leq 6$ and 80 deterministic random vectors in dimensions seven and eight, at every admissible incidence and tight-frame cosparsity. It checked 1,172 vector instances and 35,318 optimum/projection comparisons with no failure. This is not a proof; it checks the recurrence, reconstruction, zero counts, tight-frame branch, repeated-value cases, and zero-mean cases against an independent exhaustive specification.

6 Novelty check and limitations

Tillmann, Gribonval, and Pfetsch prove strong NP-hardness of general cosparsity projection and exclude an FPTAS unless $P = NP$ [2]. They explicitly cite the source oracle assumption and leave general practical approximation and further structured cases open. The source and that supporting paper already record unitary, one-dimensional finite-difference, and fused-Lasso tractable cases.

The run indexes and bounded arXiv/web searches through August 13, 2026 used combinations of complete graphs, incidence operators, graph cosparsity, tight frames, and dynamic programming. No statement of the result in this packet was found. Novelty confidence is moderate rather than exhaustive.

This packet does not give a general-operator approximation, does not address all tight frames, and does not prove new Ω -RIP bounds. Human review should focus on Lemma 1, the exact zero-count state, and the zero-DC union formula in Theorem 2.

References

- [1] R. Giryes, S. Nam, M. Elad, R. Gribonval, and M. E. Davies, *Greedy-Like Algorithms for the Cosparsity Analysis Model*, arXiv:1207.2456.
- [2] A. M. Tillmann, R. Gribonval, and M. E. Pfetsch, *Projection onto the Cosparsity Set is NP-Hard*, arXiv:1303.5305.